ECHO-7 | R5 MEASURABLE TARGETS

Source **E7070014** | Prospective acceptance specification | 05 NOV 2026

Robot: OV7670 pixels -> DE1-SoC recognition/calibration -> target qualification and motion FSM -> UGV02 UART. SW0-SW5 choose a blue bay; IR A and J3 protect motion.

Claimed grade level: PENDING - no level asserted. Current three-trial scores are pending. Historical E4/E12 observations below do not certify this revision.

Required field	T1 - Marker accuracy	T2 - Vision latency	T3 - Colour calibration	T4 - Docking gap error
Quantity + unit	Correct blue-marker identities (%) = correct / 60 expected labels per trial.	Sampled capture-to-result latency (ms): unsigned completionUs - captureStartUs.	Confirmed calibration attempt to new profile commit (ms); conservative observed upper bound.	Absolute final gap error (mm): |ruler gap -350|, measured from J3 probe front to target plane.
Test conditions	Six official blue 1-6 signs, fixed and fully visible with >=10 px margin; camera 0.40-0.70 m away. Stable indoor light, no glare. VISION_ONLY; SW8=0, SW9=0; calibrated profile held fixed.	Same scene ranges as T1, nominal 25 fps and 320 x 240 RGB565. Use complete FPGA reports, including failed results; keep build/config fixed. No newly dropped/overflow frames during a scored trial.	Red1 plus its paired blue reference, both complete and still; 0.40-0.70 m camera distance, stable indoor light, nominal 25 fps. VISION_ONLY; same build; independent rearm before each attempt.	Already-visible blue 1-4, locked within U90-150; start gap 0.60-0.70 m. Flat plastic floor, vertical board spanning camera/J3 heights, unchanged supply/load. Fixed declared runtime settings for all 3 runs.
Three-trial scoring rule	Three tutor-chosen static poses. Per pose, score the first 10 distinct reports against a photographed label/region map: 6 labels x 10. Count bad/missing/ambiguous labels as wrong; no best-frame filtering. Use the LOWEST of three rates; require zero wrong accepted IDs.	Use T1's three sets of 10 distinct reports. Take each set's maximum board-time latency, then the LARGEST of those three. A missing report or new drop/overflow fails the trial. This sampled maximum does not bound unsampled frames or camera-to-wheel latency.	Three independently rearmed attempts. Log board-time trace from confirmed attempt through eight qualified samples and new-config commit. Use the LONGEST conservative interval. Failure, missing start/commit or >2 s is not a success; ACK alone does not count.	Reset to the same start marks for 3 approaches. Stop a trial after 25 s if no autonomous DOCKED. Measure each settled ruler gap; use the LARGEST absolute error. Any timeout/manual stop/no DOCKED fails; J3 alone cannot replace the ruler.
Threshold	>=95%; zero wrong accepted IDs; every bay appears in each trial.	<=150 ms, with all three sets complete and no new drop/overflow.	All 3 commits <=2000 ms, each with 8 valid sample frames.	3/3 autonomous docks; worst absolute error <=50 mm.
Goal	100% in all three trials.	<=100 ms under the same rule.	All 3 commits <=1000 ms.	3/3 docks; worst error <=20 mm.
Predicted value	100% intended on the declared static sheet. Design expectation, not a proven physical success rate.	Provisional 100 ms budget: 40 ms capture + 60 ms processing/commit allowance. Not a proven worst-case bound.	Ideal 360 ms: (8 sample frames + 1 commit boundary) x 40 ms. Excludes pre-confirmation detection and rejected frames.	**PENDING** braking/ranging error model. A logical 350 mm stop threshold is not a physical gap prediction.
Measured value	E14: **PENDING**. E4 history: all six IDs present in 12/12 snapshots in each of three sets; this is not spatially truth-scored T1 accuracy.	E14: **PENDING**. E12 history: 38.818-57.066 ms across 90 sampled reports in one moving-target task; not the required three static sets.	E14: **PENDING**. E12 preflight showed calibrated=true and 8 sample frames; three independently timed attempts are absent.	E14: **PENDING**. Latest E12 outside-view task ended INCOMPLETE at 60.007 s, not DOCKED; no valid T4 error. Earlier partial runs cannot be joined.
Claimed grade level	**PENDING**	**PENDING**	**PENDING**	**PENDING**

Common rule: The tutor chooses within each range. Record all three trials; never select the best. T1/T2 may share captures. Use first distinct reports, retain failed/missing results and reject duplicate frame IDs. Plan about 90 s per target; T4 has a 25 s/run cap. Demonstrate IR safety before driving.

Prediction / scope: Estimates are prospective, not historical predictions. T2 excludes scene-to-UART and mechanical latency. T4 covers already-visible targets only; a complete outside-view search-to-dock demonstration remains separate and unpassed. Do not join partial runs or treat command speeds as measured wheel speeds.

Contribution / R9: T1/T2 cover the projective/anchor reader and perception pipeline; T3 covers real-pixel calibration; T4 covers target qualification and motion decisions. Repeat T1 with SW8=1 under matched conditions. If bypass also meets threshold, this target does not establish that stage's necessity.

Sources: Major_Project_Brief_2026.pdf: pp 3-4 R5/2.3 (target rules), pp 4/6 2.4/4 (level and evidence), p 7 submission. All numeric thresholds/goals are project choices: T1-T3 retained from render_member_b_datasheet.py; T4 newly declares 350 mm +/-50 mm threshold /+/-20 mm goal. Detailed provenance: docs/R5_TARGETS.md; editable specification: tools/data/echo7_r5_targets.json.